

Consistency and asymptotic normality of Maximum Likelihood Estimator (independent but not necessarily identically distributed)

Max Leung

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Newey and McFadden (1994) only includes theorems for MLE with i.i.d. assumption. I try to extend them with independent but not necessarily identically distributed assumption. The same asymptotic distribution is derived by Hoadley (1971), but not from Extremum estimator approach.

1 Consistency

Theorem 1 (Consistency of Extremum Estimator with compact parameter space)

Let $Q_0(\theta)$ be the limiting objective function and $\hat{\theta} := \operatorname{argmax}_{\theta \in \Theta} \hat{Q}_n(\theta)$ i.e., the maximizer of the sample objective function. If (i) θ_0 is the unique maximizer of $Q_0(\theta)$. (ii) $\theta_0 \in \Theta$ which is compact. (iii) $Q_0(\theta)$ is a continuous function of θ . (iv) $\sup_{\theta \in \Theta} |\hat{Q}_n(\theta) - Q_0(\theta)| \rightarrow_p 0$. Then, $\hat{\theta} \rightarrow \theta_0$.

Proof: See Theorem 2.1 of Newey and McFadden (1994).

Q.E.D.

Lemma 2 (Information Inequality; Identification Conditions for MLE)

Let $f_i(y|\mathbf{x}; \theta)$ be the conditional density and $\hat{Q}_n(\theta) = n^{-1} \sum_{i=1}^n \ln f_i(y_i|\mathbf{x}_i; \theta)$. If (i) $f_i(y|\mathbf{x}; \theta) \neq f_i(y|\mathbf{x}; \theta_0)$ for $\forall \theta \in \Theta$ and $\theta \neq \theta_0$. (ii) $\mathbb{E}[|\ln f_i(y_i|\mathbf{x}_i; \theta)|] < \infty$ for $\forall i$. (iii) $n^{-1} \sum_{i=1}^n \mathbb{E}[\ln f_i(y_i|\mathbf{x}_i; \theta)] := n^{-1} \sum_{i=1}^n Q_{0,i}(\theta) \rightarrow Q_0(\theta)$. Then, θ_0 is the unique maximizer of $Q_{0,i}(\theta)$ and thus $Q_0(\theta)$.

Proof: Replace f with f_i in the Lemma 2.2 of Newey and McFadden (1994).

Q.E.D.

Theorem 3 (Uniform LLN for independent but not necessarily identically distributed case)

If (i) $y_i|\mathbf{x}_i$ is independent but not necessarily identically distributed. (ii) parameter space Θ is compact. (iii) $a_i(y_i|\mathbf{x}_i; \theta)$ is a continuous function of θ for all $y_i|\mathbf{x}_i$ with probability one. (iv) For at least one $\delta > 0$, there exists $d_i(y_i|\mathbf{x}_i)$ such that $|a_i(y_i|\mathbf{x}_i; \theta)|^{1+\delta} \leq d_i(y_i|\mathbf{x}_i)$ for $\forall \theta \in \Theta$ and $\mathbb{E}[d_i(y_i|\mathbf{x}_i)] < \infty$. Then, $\sup_{\theta \in \Theta} |n^{-1} \sum_{i=1}^n a_i(y_i|\mathbf{x}_i; \theta) - n^{-1} \sum_{i=1}^n \mathbb{E}[a_i(y_i|\mathbf{x}_i; \theta)]| \rightarrow_p 0$ and $n^{-1} \sum_{i=1}^n \mathbb{E}[a_i(y_i|\mathbf{x}_i; \theta)]$ is a continuous function of θ .

Reference: Theorem 4.2.2 of Amemiya (1985).

Theorem 4 (Consistency of MLE with compact parameter space)

If (i) $y_i|\mathbf{x}_i$ is independent but not necessarily identically distributed with conditional density $f_i(y_i|\mathbf{x}_i; \theta_0)$. (ii) $f_i(y|\mathbf{x}; \theta) \neq f_i(y|\mathbf{x}; \theta_0)$ for $\forall \theta \in \Theta$ and $\theta \neq \theta_0$. (iii) $\theta_0 \in \Theta$ which is compact. (iv) $f_i(y_i|\mathbf{x}_i; \theta_0)$ is a continuous function of θ for all

$y_i|\mathbf{x}_i$ with probability one. (v) $\mathbb{E}[\sup_{\boldsymbol{\theta} \in \Theta} |\ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})|^{1+\delta}] < \infty$ for at least one $\delta > 0$. (vi) $n^{-1} \sum_{i=1}^n \mathbb{E}[\ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})] := n^{-1} \sum_{i=1}^n Q_{0,i}(\boldsymbol{\theta}) \rightarrow Q_0(\boldsymbol{\theta})$. Then, $\hat{\boldsymbol{\theta}} \rightarrow_p \boldsymbol{\theta}_0$.

Proof: (ii), (v) and (vi) implies Theorem 1's (i) by using Lemma 2 because (v) implies Lemma 2's (ii). (iii) is Theorem 1's (ii). Set $a_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}) = \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})$ and $d_i(y_i|\mathbf{x}_i) = \sup_{\boldsymbol{\theta} \in \Theta} |a_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})|^{1+\delta}$ in Theorem 3. (i), (iii), (iv), and (v) is or implies Theorem 3's (i), (ii), (iii), and (iv) respectively. Thus, Theorem 3 implies Theorem 1's (iv) and (iii) since $n^{-1} \sum_{i=1}^n \mathbb{E}[\ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})] \rightarrow Q_0(\boldsymbol{\theta})$ by (vi). Q.E.D.

2 Asymptotic Normality

Theorem 5 (Asymptotic Normality of Extremum Estimator)

Let $\hat{\boldsymbol{\theta}} := \operatorname{argmax}_{\boldsymbol{\theta} \in \Theta} \hat{Q}_n(\boldsymbol{\theta}) \rightarrow_p \boldsymbol{\theta}_0$. (i) $\boldsymbol{\theta}_0$ is not at the boundary of Θ . (ii) $\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \hat{Q}_n(\boldsymbol{\theta})$ exists and continuous in N where $\boldsymbol{\theta}_0 \in N$. (iii) $\sqrt{n} \nabla_{\boldsymbol{\theta}} \hat{Q}_n(\boldsymbol{\theta}_0) \rightarrow_d N(\mathbf{0}, \boldsymbol{\Sigma})$. (iv) there exists $\mathbf{H}(\boldsymbol{\theta})$ such that it is continuous at $\boldsymbol{\theta}_0$ and $\sup_{\boldsymbol{\theta} \in N} \|\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \hat{Q}_n(\boldsymbol{\theta}) - \mathbf{H}(\boldsymbol{\theta})\| \rightarrow_p 0$. (v) $\mathbf{H}(\boldsymbol{\theta}_0)^{-1}$ exists. Then, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow_d N(\mathbf{0}, \mathbf{H}^{-1} \boldsymbol{\Sigma} \mathbf{H}^{-1})$.

Proof: See Theorem 3.1 of Newey and McFadden (1994). Q.E.D.

Theorem 6 (Liapounov Central Limit Theorem)

Let y_i be independent. If $\lim_{n \rightarrow \infty} [\sum_{i=1}^n \operatorname{Var}(y_i)]^{-1/2} (\sum_{i=1}^n \mathbb{E}[|y_i - \mathbb{E}[y_i]|^3])^{1/3} = 0$. Then, $\sqrt{n}(n^{-1} \sum_{i=1}^n y_i - n^{-1} \sum_{i=1}^n \mathbb{E}[y_i]) \rightarrow_d N(0, n^{-1} \sum_{i=1}^n \operatorname{Var}(y_i))$.

Reference: Theorem 3.3.5 of Amemiya (1985).

Theorem 7 (Asymptotic Normality of MLE with compact parameter space)

If (i) $y_i|\mathbf{x}_i$ is independent but not necessarily identically distributed with conditional density $f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)$ and other assumptions in Theorem 4 holds. (ii) $\boldsymbol{\theta}_0$ is not at the boundary of Θ . (iii) $f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}) > 0$ is twice continuously differentiable in N neighborhood of $\boldsymbol{\theta}_0$. (iv) Sufficient conditions for interchanging integration and differentiation. (v) $\mathbf{J}_i = \mathbb{E}[\nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0) \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)']$ is invertible. (vi) $\mathbb{E}[\sup_{\boldsymbol{\theta} \in N} \|\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})\|^{1+\delta}] < \infty$ for at least one $\delta > 0$. (vii) $\lim_{n \rightarrow \infty} [\sum_{i=1}^n \mathbf{J}_i]^{-1/2} (\sum_{i=1}^n \mathbb{E}[\|\nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)\|^3])^{1/3} = 0$. Then, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow_d N(\mathbf{0}, (n^{-1} \sum_{i=1}^n \mathbf{J}_i)^{-1})$.

Proof: (ii) is Theorem 5's (i). $\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \hat{Q}_n(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} n^{-1} \sum_{i=1}^n \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}) = n^{-1} \sum_{i=1}^n \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})$ and thus (iii) implies Theorem 5's (ii). $\nabla_{\boldsymbol{\theta}} \hat{Q}_n(\boldsymbol{\theta}_0) = \nabla_{\boldsymbol{\theta}} n^{-1} \sum_{i=1}^n \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0) = n^{-1} \sum_{i=1}^n \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)$. $\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)] = \nabla_{\boldsymbol{\theta}} \mathbb{E}[\ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)] = \nabla_{\boldsymbol{\theta}} Q_{0,i}(\boldsymbol{\theta}_0)$ by (iv). $\nabla_{\boldsymbol{\theta}} Q_{0,i}(\boldsymbol{\theta}_0) = 0$ because Theorem 4's (ii) and (v) implies $\boldsymbol{\theta}_0$ is the unique maximizer of $\mathbb{E}[\ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})] = Q_{0,i}(\boldsymbol{\theta})$. Thus, $\operatorname{Var}(\nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)) = \mathbf{J}_i$ by (v). With (i) and (vii), we apply Liapounov Central Limit Theorem, and have $\sqrt{n} \nabla_{\boldsymbol{\theta}} \hat{Q}_n(\boldsymbol{\theta}_0) = \sqrt{n}(n^{-1} \sum_{i=1}^n \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0) - 0) \rightarrow_d N(\mathbf{0}, n^{-1} \sum_{i=1}^n \mathbf{J}_i)$. Set $a_i(\cdot) = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})$ and $d_i(\cdot) = \sup_{\boldsymbol{\theta} \in N} \|\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})\|^{1+\delta}$ in Theorem 3. Theorem 4's (i), (iii), and this theorem's (iii) and (vi) implies $\sup_{\boldsymbol{\theta} \in N} \|n^{-1} \sum_{i=1}^n \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}) - n^{-1} \sum_{i=1}^n \mathbb{E}[\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})]\| \rightarrow_p 0$ and $n^{-1} \sum_{i=1}^n \mathbf{H}_i(\boldsymbol{\theta}) := n^{-1} \sum_{i=1}^n \mathbb{E}[\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})] = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} n^{-1} \sum_{i=1}^n \mathbb{E}[\ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta})] = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} n^{-1} \sum_{i=1}^n Q_{0,i}(\boldsymbol{\theta}) \rightarrow \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} Q_0(\boldsymbol{\theta}) := \mathbf{H}(\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta}$ and thus $\mathbf{H}(\boldsymbol{\theta})$ is continuous at $\boldsymbol{\theta}_0$. (iv) implies Information Matrix equality i.e., $\mathbf{J}_i = -\mathbf{H}_i$. So, $\mathbf{J} := n^{-1} \sum_{i=1}^n \mathbf{J}_i = -n^{-1} \sum_{i=1}^n \mathbf{H}_i := -\mathbf{H}$. Therefore, (v) implies Theorem 5's (v). Finally, $\mathbf{H}^{-1} \mathbf{J} \mathbf{H}^{-1} = \mathbf{J}^{-1} \mathbf{J} \mathbf{J}^{-1} = \mathbf{J}^{-1} = (n^{-1} \sum_{i=1}^n \mathbf{J}_i)^{-1}$. Q.E.D.

Remark: It is easy to check that the conditions of Theorem 7 implies the conditions of Theorem 2 in Hoadley (1971) for the asymptotic normality. For instance, N5 in Hoadley (1971) is $\mathbb{E}[\nabla_{\theta} \ln f_i(y_i|\mathbf{x}_i; \boldsymbol{\theta}_0)] = 0$ in our proof. N6 is Information Matrix equality, which is implied by (iv) of Theorem 7. N7 is $-n^{-1} \sum_{i=1}^n \mathbf{H}_i(\boldsymbol{\theta}) \rightarrow -\mathbf{H}(\boldsymbol{\theta})$ in our proof. N8 is Liapounov condition, which is Theorem 7's (vii). N9 is dominance condition, which is (vi) of Theorem 7. The asymptotic distribution is exactly the same. However, the conditions in Theorem 7 (I believe) is more primitive i.e., easier to interpret.

3 Application

Binary outcome model with $Pr(y_i = 1|\mathbf{x}_i) = F_i(\mathbf{x}'_i\boldsymbol{\beta}_0)$

$y_i|\mathbf{x}_i$ is independent and Bernoulli distributed with parameter $p_i = Pr(y_i = 1|\mathbf{x}_i) = F_i(\mathbf{x}'_i\boldsymbol{\beta}_0)$. Thus, it is not identically distributed since different i can have different $F_i(\cdot)$ and thus p_i . By specifying conditions which implies the conditions in Theorem 4 and Theorem 7, $\hat{\boldsymbol{\beta}}$ is consistent and asymptotic normal. In practice, $F_i(\cdot)$ is usually chosen to be Logistic function or standard normal distribution function for $\forall i$. This model relaxes this assumption. From the latent variable approach of binary outcome model, $F_i(\cdot)$ is the distribution function of the error of latent variable y_i^* e.g., utility level for making decision. It is natural to assume different individual has different distribution of random utility.

For example, we can extend Robit model (Liu, 2004) to heterogeneous Robit model $F_{t,\nu_i}(\mathbf{x}'_i\boldsymbol{\beta}_0) = \int_{-\infty}^{\mathbf{x}'_i\boldsymbol{\beta}_0} t(z;\nu_i)dz$ where $t(\cdot)$ is the density of standard student's t random variable, ν_i is its parameter. When $\nu_i \rightarrow \infty$, $t(\cdot) \rightarrow \phi(\cdot)$ and thus $F_{t,\nu_i}(\cdot) \rightarrow \Phi(\cdot)$ i.e., Probit model. When ν_i is close to 9, $F_{t,\nu_i}(\cdot)$ is close to *Logistic*($\cdot$) i.e., Logit model / Logistic regression. If ν_i s are estimated simultaneously with $\boldsymbol{\beta}$, the data select the choice between Probit, Logit or mixture of them for each i within heterogeneous Robit model.

4 Reference

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